Statistical Inference and Modeling

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Lectures

- Lecture 1: Introduction
- Lecture 2: Model selection
- Lecture 3: Resampling
- Lecture 4: Linear regression
- Lecture 5: Robust regression
- Lecture 6: Classification
- Lecture 7: Logistic regression
- Lecture 8: Neural networks
- Lecture 9: K-nearest neighbor
- Lecture 10: Bayesian decision, LDA, QDA, and naive Bayes
- Lecture 11: Support vector machine
- Lecture 12: Tree-based methods
- Lecture 13: Bagging, random forecast, boosting
- Lecture 14: Unsupervised learning and clustering algorithms
- Lecture 15: Unsupervised learning and PCA
- Lecture 16: Gaussian mixture model clustering
- Lecture 17: DBSCAN clustering

7 ~~8~~. Logistic regression

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$\longrightarrow$ formulation, optimization problem.

Binary Classification.

Consider the problem of classifying data into two classes: $y \in \{0, 1\}$

$x = [x_1 \ x_2 \ \dots \ x_n]$

Goal: find a model that provides $p(y = 1|\boldsymbol{x} = [x_i]_{i=1:n})$ or $p(y = 0|\boldsymbol{x} = [x_i]_{i=1:n})$.

Settings: We a training dataset $\{(\boldsymbol{x}_j, y_j)\}_{j=1}^{m}$ where m is the number of samples, y_j is the label and $\boldsymbol{x}_j$ is the input feature.

Example:

- An input feature $\boldsymbol{x} = [x_1, x_2, x_2, x_4]$ contains 4 features, *i.e.*, LDL, BMI, alcohol, age.
- The goal is to predict whether the patient has heart disease; $y = 1$ is 'yes' and $y = 0$ is 'no'.
- $p(\text{heart disease} = \text{no}|\ \text{LDL} = 70\text{mg/dL},\ \text{BMI} = 20, \text{alcohol} = 10\text{mg/dL},\ \text{age} = 20)$

Logistic regression

$x^T\beta$ ← Input features

linear model parameter

A logistic function is used to give output between 0 and 1:

$$f(x) = \frac{1}{1+\exp -x} = \frac{\exp x}{1+\exp x}$$

the probabilistic value

has S-shape

This is a normal form of logistic, aka. sigmoid function. A logistic model uses the logistic function to explain Y from predictors through:

$$p(y=1|\boldsymbol{x}) = \frac{\exp \boldsymbol{x}^T\boldsymbol{\beta}}{1+\exp \boldsymbol{x}^T\boldsymbol{\beta}},$$

$$p(y=0|\boldsymbol{x}) = \frac{1}{1+\exp \boldsymbol{x}^T\boldsymbol{\beta}}$$ (which is from $1-p(y=1|\boldsymbol{x})$)

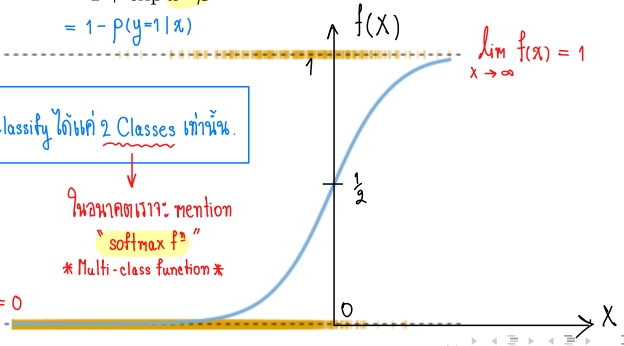
$= 1 - p(y=1|x)$

$\boxed{f(x) = \frac{e^x}{1+e^x}}$ สามารถ Classify ได้แค่ 2 Classes เท่านั้น.

ในอนาคตเราจะ mention

"softmax f^{n}"

* Multi-class function *



$\lim_{x\to-\infty} f(x) = 0$

Logistic regression

Problem: fitting the logistic model

$$P(Y=y|\boldsymbol{x},\ \boldsymbol{\beta}) = \begin{cases} p(y=1|\boldsymbol{x}) = \frac{\exp \boldsymbol{x}^T\boldsymbol{\beta}}{1+\exp \boldsymbol{x}^T\boldsymbol{\beta}} & \text{if } y=1 \\ p(y=0|\boldsymbol{x}) = 1 - p(y=1|\boldsymbol{x}) = 1 - \frac{\exp \boldsymbol{x}_j^T\boldsymbol{\beta}}{1+\exp \boldsymbol{x}^T\boldsymbol{\beta}} & \text{otherwise} \end{cases}$$

from a training data set $\{(y_j, \boldsymbol{x}_j)\}_{j=1}^m$ to learn the model parameters β.

- Note that the exponential term of $p(y=1|\boldsymbol{x})$ is a linear function of the model parameters: $\boldsymbol{x}^T\boldsymbol{\beta} = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_p$.
- This mean that we can apply some parts of the learning in linear regression to the logistic regression.
- In fact, we will learn the model parameters β through solving a maximum likelihood problem.

เมื่อ $P(Y=1|X) = 0.5 \longleftrightarrow X^T\beta = 0$; $\hat{y} = \begin{cases} 1, & \boxed{X^T\beta > 0} \\ 0, & \boxed{X^T\beta < 0} \end{cases}$ $\rightarrow$ เราจะเรียกว่า $X^T\beta$ เป็น "decision boundary"

Derivation of log-likelihood for learning the parameters

Suppose $\{(y_j, \boldsymbol{x}_j)\}_{j=1}^m$ are available where $y_j = 0, 1$.

- We can write ...

$$P(Y = y|\boldsymbol{x},\ \boldsymbol{\beta}) = \begin{cases} p(y=1|\boldsymbol{x}) = \frac{\exp \boldsymbol{x}^T\boldsymbol{\beta}}{1+\exp \boldsymbol{x}^T\boldsymbol{\beta}} & \text{if } y = 1 \\ 1 - p(y=1|\boldsymbol{x}) = \frac{1}{1+\exp \boldsymbol{x}^T\boldsymbol{\beta}} & \text{otherwise} \end{cases}$$

with the following $P(Y = y|\boldsymbol{x},\ \boldsymbol{\beta}) = p(y=1|\boldsymbol{x})^y(1-p(y=1|\boldsymbol{x}))^{1-y}$. ⇒ Bernouli

- For m independent samples: $\mathcal{L}(\boldsymbol{\beta}) = \prod_{j=1}^m P(Y = y_j|\boldsymbol{x}_j,\ \boldsymbol{\beta})$

$$\log \mathcal{L}(\boldsymbol{\beta}) = \log \prod_{j=1}^m P(Y = y_j|\boldsymbol{x}_j,\ \boldsymbol{\beta})$$

← log-likelihood

goal $\underset{\beta}{\operatorname{argmax}}\ \mathcal{L}(\beta) = \underset{\beta}{\operatorname{argmax}}\ p(y_1, y_2, \ldots, y_m | X; \beta)$

$$= \sum_{j=1}^m \log P(Y = y_j|\boldsymbol{x}_j,\ \boldsymbol{\beta})$$

$$= \sum_{j=1}^m \left[y_j \log p(y=1|\boldsymbol{x}_j;\ \boldsymbol{\beta}) + (1-y_j)\log(1-p(y=1|\boldsymbol{x}_j;\ \boldsymbol{\beta})) \right]$$

$$= \sum_{j=1}^m \left[y_j \log\left(\frac{\exp \boldsymbol{x}_j^T\boldsymbol{\beta}}{1+\exp \boldsymbol{x}_j^T\boldsymbol{\beta}}\right) + (1-y_j)\log\left(\frac{1}{1+\exp \boldsymbol{x}_j^T\boldsymbol{\beta}}\right) \right]$$

$[m] = \{1, 2, 3, \dots, m\}$.

These $j \in [m]$ can be separated into $P = \{j \in [m] \ \& \ y_j = 1\}$ and $Q = \{j \ \in [m] \& \ y_j = 0\}$ where $m = |P| + |Q|$.

$y_j = 1 \ ; \forall j \in P$ $\quad$ $y_j = 0 \ ; \forall j \in Q$

$$\Rightarrow \log \mathcal{L}(\boldsymbol{\beta}) = \underline{\sum_{p \in P} \left(\boldsymbol{x}_j^T \boldsymbol{\beta} - \log\left(1 + \exp \boldsymbol{x}_j^T \boldsymbol{\beta}\right)\right)} - \underline{\sum_{q \in Q} \log\left(1 + \exp \boldsymbol{x}_q^T \boldsymbol{\beta}\right)}$$

$$= \sum_{p \in P} \boldsymbol{x}_j^T \boldsymbol{\beta} - \sum_{j \in P+Q} \log\left(1 + \exp \boldsymbol{x}_j^T \boldsymbol{\beta}\right)$$

$j \in \{1, 2, 3, \dots, m\}$

One can estimate for β by maximizing the **log-likelihood**

$$\boldsymbol{\beta}^* = \underset{\boldsymbol{\beta}}{\mathbf{argmax}} \ \log \mathcal{L}(\boldsymbol{\beta})$$

Linear f^n $\quad$ Non-Linear f^n

$$\beta^* = \underset{\boldsymbol{\beta}}{\mathbf{argmax}} \ \boxed{\sum_{p \in P} \boldsymbol{x}_p^T \boldsymbol{\beta}} - \boxed{\sum_{j=1}^{m} \log\left(1 + \exp \boldsymbol{x}_j^T \boldsymbol{\beta}\right)}$$

This is non-linear unconstrained optimization problem (can be solved by Newton/ Quasi-Newton)

ถ้า LP ก็ใช้ Simplex Method ได้

Default credit card payment

$$p(Y=1|X) = \frac{\exp(\beta_0+\beta_1 X)}{1+\exp(\beta_0+\beta_1 X)}$$

$Y = \begin{cases} 1 \rightarrow \text{บัตรโดนอายัติ} \\ 0 \rightarrow \text{บัตรไม่โดน} \end{cases}$

Example: logistic regression for the data on credit card payment.

Use $\hat{\beta}$ from the table we can make an estimate of Y
($Y = 1$ the credit card is defaulted; $Y = 0$ the credit card is not defaulted)

	Coeff. $\hat{\beta}_i$	Std. Err	Z-statistic	P-value
β_1 = Student [Y/N]	0.4049	0.1150	3.52	0.0004
β_0 = Intercept	-3.5041	0.0707	-49.55	< 0.0001

What is the Intercept?

$$\boxed{P(Y=1|X=x) = \frac{\exp^{\beta_0+\beta_1\times x}}{1+\exp^{\beta_0+\beta_1\times x}} = \frac{\exp^{-3.5041+0.4049\times x}}{1+\exp^{-3.5041+0.4049\times x}}}$$

$$P(Y=1|X=\underbrace{\text{not student}}_{=0}) = \frac{\exp^{-3.5041+0.4049\times 0}}{1+\exp^{-3.5041+0.4049\times 0}} = 0.0292 \rightarrow P(Y=0|X=\text{student}) = 1-0.0298$$

$$P(Y=1|X=\underbrace{\text{is a student}}_{=1}) = \frac{\exp^{-3.5041+0.4049\times 1}}{1+\exp^{-3.5041+0.4049\times 1}} = 0.0431$$

$P(Y=0|X=\text{student}) = 1-0.0431$

Example: logistic regression for the data on credit card payment.

Use $\hat{\beta}$ from the table we can make an estimate of Y
(Y : whether or not the credit card is likely to be defaulted.)

P-value $= 2 \times P(Z > |Z_{ob}|)$

	Coeff. $\hat{\beta}_i$	Std. Err	Z-statistic	P-value
Student [Y/N]	0.4049	0.1150	3.52	0.0004
Intercept	-3.5041	0.0707	-49.55	< 0.0001

Standard Error is the standard deviation of β. Let $\hat{\beta} = \mathrm{E}[\beta_i]$. → Unbiasness.

$$\mathrm{Var}[\beta_i] = \mathrm{E}[||\beta_i - \hat{\beta}||_2^2]$$

$\hat{\beta} = \frac{\hat{\beta}_0 + \hat{\beta}_1}{2} = \frac{1}{n+1}\sum_{i=0}^{n} \hat{\beta}_i$

Z-statistic measures the deviation from the null hypothesis $\beta_i = 0$

$$z = \frac{(x - \cancel{H_0}\,0)}{\sigma}$$

; $H_0 : \beta_i = 0$ vs $H_1 : \beta_i \neq 0$; $\forall i$

P-value for the null hypothesis $\beta_i = 0$.

- The smaller P-value = the more likely the null hypothesis to be rejected. reject if p-value $< \alpha$.
- For example, β_0 is unlikely a zero value (or it is likely to be non-zero, which means that its value is important).

Using one predictor (student status) versus three predictors.

	Coeff. $\hat{\beta}_i$	Std. Err	Z-statistic	P-value
Student [Y/N]	0.4049	0.1150	3.52	0.0004
Intercept	-3.5041	0.0707	-49.55	< 0.0001

	Coeff. $\hat{\beta}_i$	Std. Err	Z-statistic	P-value
Balance	0.0057	0.0002	24.74	< 0.0001
Income	0.0030	0.0082	0.37	0.7115
Student [Y/N]	-0.6468	0.2362	-2.74	0.0062
Intercept	-10.8690	0.4923	-22.08	< 0.0001

- Left Table: Positive coefficient of student indicates that students are more likely to have a defaulted credit card than non-students. ส.ป.ส บวก
- Right Table: Negative coefficient of student status suggests the opposite. ส.ป.ส. ลบ

How to read these two tables? ...

Correlated predictors

	Coeff. $\hat{\beta}_i$	Std. Err	Z-statistic	P-value
Student [Y/N]	0.4049	0.1150	3.52	0.0004
Intercept	-3.5041	0.0707	-49.55	< 0.0001

	Coeff. $\hat{\beta}_i$	Std. Err	Z-statistic	P-value
Balance	0.0057	0.0002	24.74	< 0.0001
Income	0.0030	0.0082	0.37	0.7115
Student [Y/N]	-0.6468	0.2362	-2.74	0.0062
Intercept	-10.8690	0.4923	-22.08	< 0.0001

- Left Table: Positive coefficient of student indicates that students are more likely to have a defaulted credit card than non-students. .
- Right Table: Negative coefficient of student status suggests the opposite.

How to read these two tables?

- Left Table: *for a fixed value of intercept, a student is more likely to default than a non-student.*
- Right Table: *for a fixed value of balance and income, a student is less likely to default than a non-student*, or

Balance Income Student

student	$p(Y=1 \mid X=(1500, 40000, 1)) = 0.068$
non-student	$p(Y=1 \mid X=(1500, 40000, 0)) = 0.105$

การเพิ่มค่า Student
จาก 0 → 1
ทำให้ prob ลดลง $\frac{0.105}{0.068}$ เท่า

This means Balance & Income ⇒ Chance of having credit card default
Can we check?

$$\boxed{\text{Odds Ratio}_i = e^{\beta_i}}$$

$$\ln(Odds) = -5 + 0.02(Score)$$

2. การหาค่า Odds Ratio (OR)

ค่า $\beta_1 = 0.02$ นั้นอ่านยากเพราะอยู่ในรูปของ Log เพื่อให้เป็น Odds Ratio เราต้องกำจัด Log ออกโดยการใช้ Exponential (e) ครับ

$$OR = e^{\beta_1} = e^{0.02}$$

$$OR \approx 1.0202$$

$$\boxed{\text{Odds Ratio}_i = e^{\beta_i}}$$

3. การตีความหมาย

จากค่า $\boxed{OR \approx 1.02}$ เราจะตีความในเชิงธุรกิจได้ว่า:

- หากลูกค้ามี **คะแนนเครดิตเพิ่มขึ้น 1 หน่วย**
- **Odds (แต้มต่อ)** ในการที่จะได้รับอนุมัติสินเชื่อจะ **เพิ่มขึ้นเป็น 1.02 เท่า** (หรือเพิ่มขึ้นประมาณ 2%)

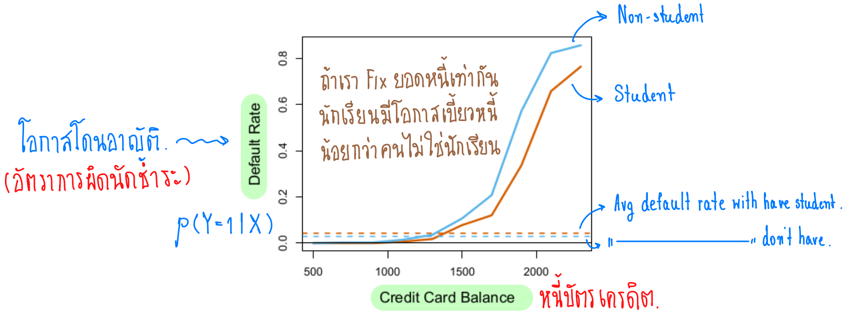
หมายเหตุ: หากตัวแปรอิสระเป็นแบบกลุ่ม (เช่น เพศชาย/หญิง) ค่า OR จะเป็นการเปรียบเทียบระหว่างกลุ่มโดยตรง แต่ถ้าเป็นตัวแปรต่อเนื่องแบบคะแนนเครดิต ค่า OR จะหมายถึงการเปลี่ยนแปลงต่อ "1 หน่วย" ที่เพิ่มขึ้นครับ

4. สรุปความสัมพันธ์ในรูปเลขยกกำลัง

ถ้าเราต้องการหา Odds ของใครคนใดคนหนึ่งโดยเฉพาะ เราสามารถเปลี่ยนสมการให้อยู่ในรูปนี้ได้เลย:

$$Odds = e^{\beta_0 + \beta_1 X} = e^{-5} \cdot (1.02)^{Score}$$

จะเห็นว่าทุกๆ Score ที่เพิ่มขึ้น 1 แต้ม จะกลายเป็นตัวคูณ 1.02 เข้าไปเรื่อยๆ นั่นเองครับ

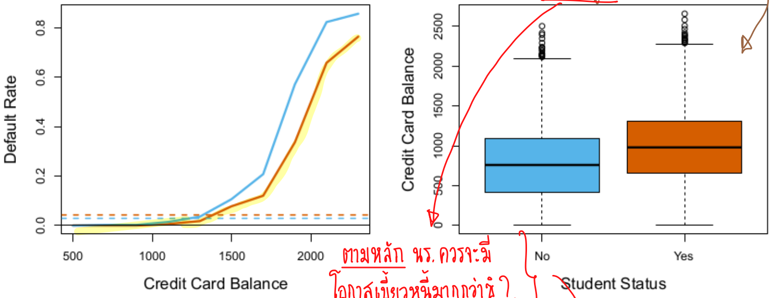


Orange color denotes students; blue color denotes non-students

Note. Higher default rate == higher chance of having credit card default!

- LHS. Two curves plots the default rates vs. credit balance.
  At each point of the credit balance, the orange line (students) is lower than blue one (non-student).
  ⇒ **An individual student** with a given balance tends to have a lower chance of default (lower default rate).
- LHS. In the plots there are two dash lines, which gives a **contradict information.** The dash lines provides the average default rate.
  ⇒ Here, the students' default rate is higher than the non-students'!
  Luckily we have the box plot to explain this!

จาก box plot แสดงว่าโดยส่วนมากแล้ว
นักเรียนจะมียอดหนี้สูงกว่าคนที่ไม่ใช่นักเรียน.

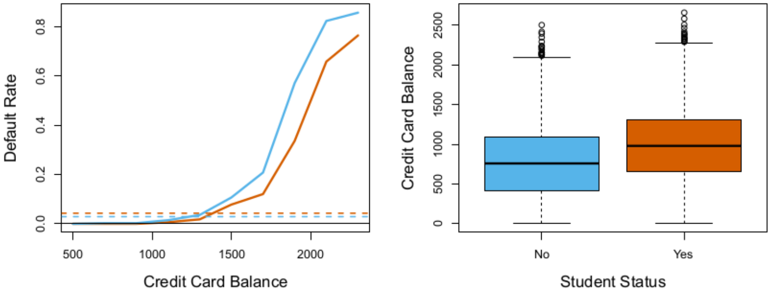


ตามหลัก นร. ควรจะมี
โอกาสเบี้ยวหนี้มากกว่ารึ ?

Orange color denotes students; blue color denotes non-students

- Box plot ⇒ 'student status' and 'balance' are correlated.
- RHS. The box plots suggest that the students as a whole have the higher credit card balance. The non-students have the lower credit card balance.
- This reveals the underlie fact that students, **as a whole**, tend to have higher credit card balance, which further tend to have a higher default rate.

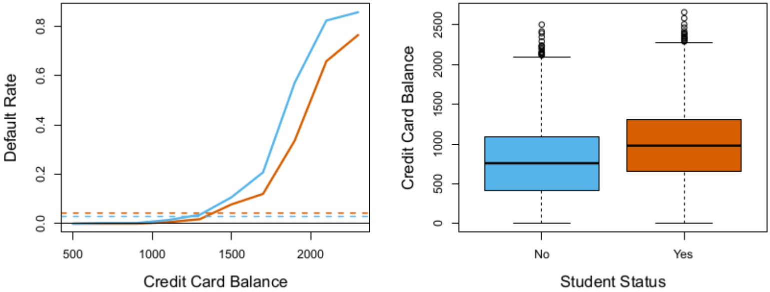
Confounding.



Orange color denotes students; blue color denotes non-students

From this information, if you were a bank clerk....

- At a given balance, whose card will less likely to be defaulted: *the student's or the non-student's* ?
  Answer ⇒ the student's.
  → ถ้าเราว่ามีคนมี balance ⇒ ใครจะมีโอกาส default rate ต่ำกว่า.
- If you give the credit to a student, what should you be careful about?
  Answer ⇒ That they maybe using the money from one credit card to pay another. So, their credit balance will be seemingly high, until one day they cannot effort to pay the entire loop.
  → ถ้าเป็นนิสิต ต้อง concern อะไรมั้ย ?.



Orange color denotes students; blue color denotes non-students

What else?

A confounding problem: a result obtained from one predictor is different from using multiple predictors when there is correlation among the predictors.

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[1] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning*. Springer Series in Statistics. New York, NY, USA: Springer New York Inc., 2001.

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